Statistic 403/603

Mid-term exam 2

NAME:

STUDENT NUMBER:

SIGNATURE:

INSTRUCTIONS:

1. Do not panic. It rarely helps.
2. Read all questions carefully.
3. Calculators are permitted (none that have communication ability).
4. Where computations are required, you should show work.
5. If a question calls for an explanation, provide one that is complete.
6. Cross out any material you do not wish to have considered.
7. **Correct answers with insufficient justification or accompanied by additional incorrect statements will not receive full credit.**
8. Correct/incorrect answers without clearly displayed work cannot be considered for partial credit.
9. You may lose marks if your explanations are incomplete, missing or poorly worded.

In accordance with the SFU Academic Honesty Policy (T 10.02), academic dishonesty in any form will not be tolerated. Prohibited acts include, but are not limited to, the following:

- making use of any books, papers or electronic devices that allow communication,
- speaking or communicating with other students who are writing exams,
- copying the work of other students or purposely exposing written papers to the view of other students

Formulas:

Simple random sampling (SRS)

Sample mean: $\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n}$

Sample standard deviation: $s = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1}}$

Standard error of the mean: $se(\bar{y}) = \sqrt{(1 - \frac{n}{N})\frac{S^2}{n}}$

Estimate of the population total: $\hat{\tau} = N\bar{y}$

Standard error for the estimate of the population total: $se(\hat{\tau}) = N\sqrt{(1 - \frac{n}{N})\frac{S^2}{n}}$

Standard error for the estimate of the population proportion: $se(\hat{p}) = \sqrt{(1 - \frac{n}{N})\frac{P(1-P)}{n-1}}$

Sample size for SRS: $n \geq \frac{z_{\alpha/2}^2 s^2}{e^2 + \frac{z_{\alpha/2}^2 s^2}{N}}$

Stratified random sampling:

Estimate of the population mean (stratified random sampling): $\hat{\mu}_{str} = \bar{y}_{str} = \sum_{h=1}^{H} \frac{N_h}{N}\bar{y}_h = \sum_{h=1}^{H} W_h \bar{y}_h$

Standard error of stratified random sampling estimate of the pop. mean: $se(\mu_{str}) = \sqrt{\sum_{h=1}^{H} W_h^2 \frac{S_h^2}{n_h}(1 - n_h/N_h)}$

Cluster sampling:

Cluster sampling estimate of population mean: $\hat{\mu} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}$

Standard error of the cluster sampling estimator of the pop. mean: $se(\hat{\mu}) = \sqrt{\frac{1}{\bar{M}^2}\frac{S_{dif}^2}{n}(1 - n/N)}$

Cluster sampling estimator of pop. total: $\hat{\tau} = \hat{\mu}M$

Cluster sampling standard error for the estimator of the population total: $se(\hat{\tau}) = se(\hat{\mu})M$

1. (6 marks) A professor in the Biology Department is conducting a study on species diversity and stability. The professor constructs 9 representative systems that are similar to the larger environmental system in the real-world (these are called *microcosms*). To each microcosm, the professor randomly assigns three of each to a low, medium, and high diversity treatment (depending upon the number of plant and animal species included). These are grown in 3 growth chambers at constant temperatures for a year, and the treatments are randomly assigned within each chamber. The productivity and species richness at the end of the experiment are measured. The resulting arrangement of microcosms in the chambers was as follows:

Chamber 1	**Chamber 2**	**Chamber 3**
H	M	H
L	L	M
M	H	L

a. The professor analyzed the data using a one-way ANOVA testing for differences between the treatments. Explain why is this a problem?

b. Write out the appropriate statistical model for this experiment, listing the appropriate assumptions.

$$y_{ij} = \mu_i + block_j + \epsilon_{ij}$$

2. The following quote was found in a scientific paper: "This experiment had 83% power to detect a difference in 3 kg in the mean growth among the treatments. The observed difference in the means was only 0.5 kg. Consequently, I am confident the null hypothesis was true."

a. Explain the first statement about power in terms that a non-statistician can understand.

b. Do you agree with the last sentence? Why or why not?

3. According to Wikipedia, *Turbidity* is the cloudiness or haziness of a liquid caused by small particles that are generally invisible to the naked eye. The measurement of **turbidity** is a key test of water quality.

   Consider an experiment that aims to measure the impact of clear cutting forests upon the *turbidity* of streams and possible consequences on fish health. If the turbidity caused by clear cutting forests impacts fish health, the Department of Oceans and Fisheries will impose stricter environmental standards. On the other hand, if no impact on fish heath is found, nothing will be done.

   Steidhl et al. (1997) recommends against the uncritical use of hypothesis testing. In particular, he was disturbed by the failure to recognize the costs of Type I and Type II errors.

   Explain in non- technical language what these errors are (Type I and Type II errors), and the consequences of these errors in the context of experiments to measure the impact of clear cutting upon turbidity of streams and possible consequences on fish health.

4. The yields of 2 varieties of corn (C1 and C2) and 2 fertilizers (F1 and F2) are to be compared using a greenhouse study. Within the greenhouse, there are 32 tables available for use arranged in a rectangular pattern of 8 by 4 tables. Eight replications of each treatment are to be done. The treatments are to be assigned to the tables where the corn is grown. After growth, the yields of each table are recorded (in kilograms).

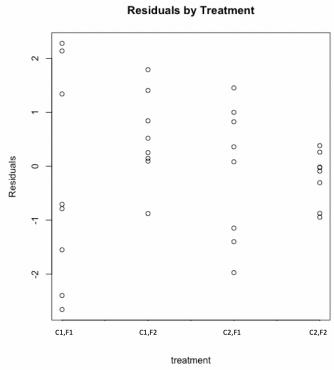
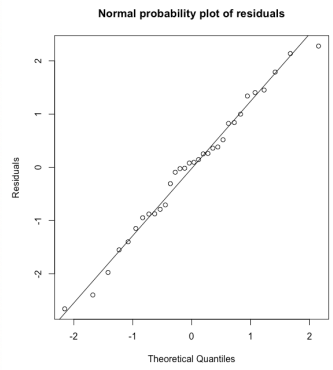
    a. Show an experimental layout that uses a completely randomized design to investigate the effect of the two factors. What are the main advantages and disadvantages of this design?

    b. There is heating vent near one of the short sides of the rectangular pattern of 8 by 4 tables. The scientists feel that there may be an impact on the yield of being close to the vent. Show an experimental layout that uses a randomized block design to investigate the effect of the two factors. What are the main advantages of this design in comparison to the design in part a?

c. It turns out that, due to supply chain issues, the corn seed is only sold in small quantity bags (i.e., there is not a lot of seed in the bags). Fortunately, our resourceful scientists are able to get enough seed for their experiments by purchasing many smaller bags of corn seed from several suppliers. Unfortunately, each bag of seed contains only enough for two trials. To make matters worse, there is quite a bit of variation from bag to bag. Taking into account the air vent in part b., show an experimental layout that uses a split-plot design with main plots as a randomized complete block design. Write out the statistical model for this experiment.

d. Suppose that the experiment in part a. was performed (ignore parts b. and c.). The following table shows the mean responses for each of the four treatments.

		Corn variety	
		C1	**C2**
Fertilizer type	**F1**	10	15
	F2	20	5

Draw an interaction plot (or profile plot) for these results (label all axes and lines). Comment on whether or not you would expect to see a significant interaction effect.

e. Following the experiment, the scientists fit a 2-factor ANOVA model for the experiment. Before continuing with their analysis, they decided to investigate the residuals. Below are some plots of the residuals. Comment on whether or not any of the model assumptions appear to be violated.